

$$28 > \frac{0.70}{0.09 - 0.045} = 15.56.$$

Coca-Cola's forward price-to-earnings ratio of 28 is much greater than 15.56, which is computed from the equation. Investor expectations of cash flow growth and return are inconsistent with Coca-Cola's forward price to earnings ratio. Specifically, an investor should consider a short position in Coca-Cola stock in the belief that its price should decline because its current price to earnings ratio is well above what its fundamentals imply. As shown in results for questions 1 and 2 in this example, expectations for the required return and growth rate must be such that $r - g = 2.50\%$ to justify a forward price-to-earnings ratio of 28 given the expected dividend payout ratio of 70 percent.

CASH FLOW ADDITIVITY

4



explain the cash flow additivity principle, its importance for the no-arbitrage condition, and its use in calculating implied forward interest rates, forward exchange rates, and option values

The time value of money trade-off between cash flows occurring today versus those in the future may be extended beyond pricing future cash flows today or calculating the implied return on a single instrument using the **cash flow additivity principle**. Under cash flow additivity, the present value of any future cash flow stream indexed at the same point equals the sum of the present values of the cash flows. This principle is important in ensuring that market prices reflect the condition of no arbitrage, or that no possibility exists to earn a riskless profit in the absence of transaction costs.

Let's begin with a basic example to demonstrate the cash flow additivity principle.

EXAMPLE 12

Basic Cash Flow Additivity

Let's assume that you have GBP100 to invest, and have two strategies from which to choose with the following cash flow streams as shown in Exhibit 9.